Tech Bite #10: Modes and Bandwidth -- FM, SSB, CW

The ways a radio carries information -- FM, SSB, and CW -- and the one number that separates them: bandwidth.

Group T8A · 12 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The Big Picture

A **mode** is the method a radio uses to carry information. This Bite covers the modes a Technician meets most – **FM**, **SSB**, and **CW** – and one number that separates them: **bandwidth**, the slice of spectrum a signal occupies.

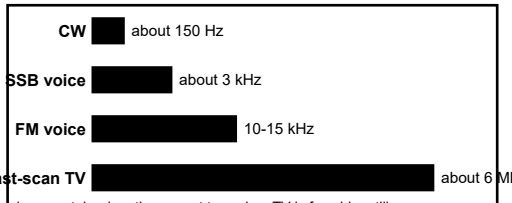
The Three Main Modes

- **FM** (frequency modulation) is the everyday mode for **VHF/UHF voice repeaters**. (T8A04) Its weakness compared to SSB: only one signal can be received at a time – the strongest one captures the receiver. (T8A12) VHF **packet radio** is also sent using FM or PM. (T8A02)
- **SSB** (single sideband) is a form of **amplitude modulation**. (T8A01) It is the mode for **long-distance, weak-signal** work on VHF and UHF (T8A03), because it has a much **narrower bandwidth** than FM. (T8A07) On 10 meters, VHF, and UHF, SSB normally uses the **upper sideband** (USB). (T8A06)
- **CW** (Morse code) has the **narrowest bandwidth** of any mode. (T8A05)

Bandwidth: How Much Spectrum a Signal Uses

Bandwidth is the width of the slice of spectrum a signal fills. Narrower signals are more efficient – more of them fit in a band. From narrowest to widest:

- **CW**: about 150 Hz (T8A11)
- **SSB voice**: about 3 kHz (T8A08)
- **FM voice** (VHF repeater): 10 to 15 kHz (T8A09)
- **AM fast-scan TV**: about 6 MHz (T8A10)



Order is correct; bar lengths are not to scale – TV is far wider still.

Bandwidth, narrowest to widest. CW sips spectrum; TV floods it.

Mental Cheat Sheet

- **FM/PM**: VHF/UHF voice repeaters, and VHF packet.
- **SSB** is a form of **amplitude modulation**; it uses **USB** on 10 m and up.
- **SSB** is the weak-signal, long-distance VHF/UHF voice mode.
- **CW** has the **narrowest** bandwidth of all.
- Bandwidths: CW 150 Hz, SSB 3 kHz, FM 10-15 kHz, fast-scan TV 6 MHz.
- FM's drawback: only the strongest signal is received at a time.

Common Traps

- **SSB** is amplitude modulation – not a form of FM.
- **CW** is the narrowest mode; SSB is narrower than FM but not the narrowest.
- SSB on 10 m / VHF / UHF uses **upper** sideband.
- FM voice on a repeater is **10-15 kHz** – not 150 kHz.
- Fast-scan TV is about **6 MHz** wide – vastly wider than any voice mode.
- FM's disadvantage: only **one signal at a time**, not "poorer voice quality."

Takeaway: The three modes a Technician uses most: **FM** for repeaters and local voice, **SSB** (a form of AM, upper sideband) for weak-signal distance work, and **CW** for the narrowest signal of all. **Bandwidth** is how much spectrum a signal uses – CW about 150 Hz, SSB about 3 kHz, FM 10-15 kHz, and fast-scan TV about 6 MHz. Narrower signals share the band better.

Bite 10 Question Bank (T8A)

T8A01. Which of the following is a form of amplitude modulation?
A. Spread spectrum
B. Packet radio
C. Single sideband
D. Phase shift keying (PSK)

T8A02. What type of modulation is commonly used for VHF packet radio transmissions?
A. FM or PM
B. SSB
C. AM
D. PSK

T8A03. Which type of voice mode is often used for long-distance (weak signal) contacts on the VHF and UHF bands?
A. FM
B. DRM
C. SSB
D. PM

T8A04. Which type of modulation is commonly used for VHF and UHF voice repeaters?
A. AM
B. SSB
C. PSK
D. FM or PM

T8A05. Which of the following signal types has the narrowest bandwidth?
A. FM voice
B. SSB voice
C. CW
D. Slow-scan TV

T8A06. Which sideband is normally used for 10-meter HF, VHF, and UHF single-sideband communications?
A. Upper sideband
B. Lower sideband
C. Suppressed sideband
D. Inverted sideband

T8A07. What is one characteristic of single sideband (SSB) compared to FM?
A. SSB signals are easier to tune in correctly
B. SSB signals are less susceptible to interference
C. SSB signals have narrower bandwidth
D. SSB signals are less susceptible to high SWR

T8A08. What is the approximate bandwidth of a typical single sideband (SSB) voice signal?
A. 1 kHz
B. 3 kHz
C. 6 kHz
D. 15 kHz

T8A09. What is the approximate bandwidth of an FM voice signal on VHF repeaters?
A. Less than 500 Hz
B. About 150 kHz
C. Between 10 and 15 kHz
D. Between 50 and 125 kHz

T8A10. What is the approximate bandwidth of AM fast-scan TV transmissions?
A. More than 10 MHz
B. About 6 MHz
C. About 3 MHz
D. About 1 MHz

T8A11. What is the approximate bandwidth required to transmit a CW signal?
A. 2.4 kHz
B. 150 Hz
C. 1000 Hz
D. 15 kHz

T8A12. Which of the following is a disadvantage of FM compared with single sideband?
A. Voice quality is poorer
B. Only one signal can be received at a time
C. FM signals are harder to tune
D. FM signals are more susceptible to high SWR

Answer Key

T8A01: C T8A02: A T8A03: C T8A04: D

T8A05: C T8A06: A T8A07: C T8A08: B

T8A09: C T8A10: B T8A11: B T8A12: B